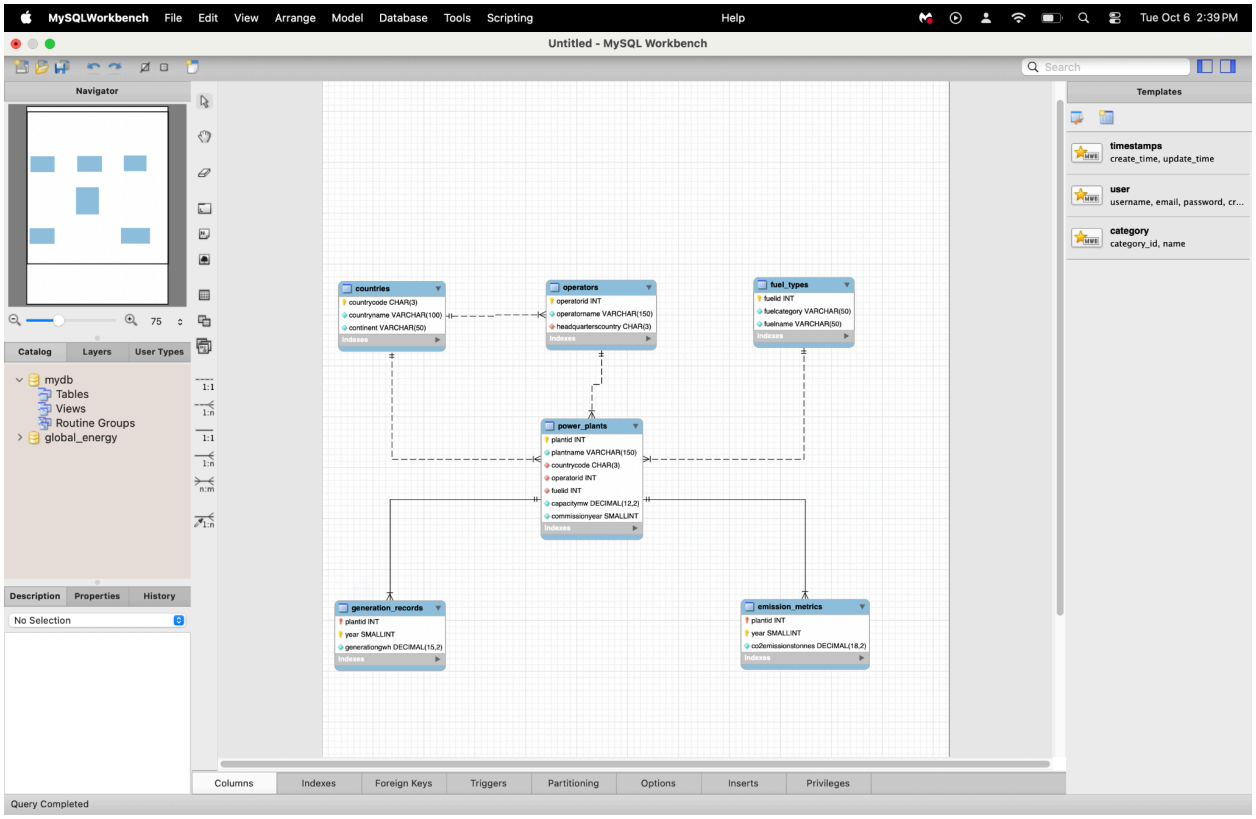
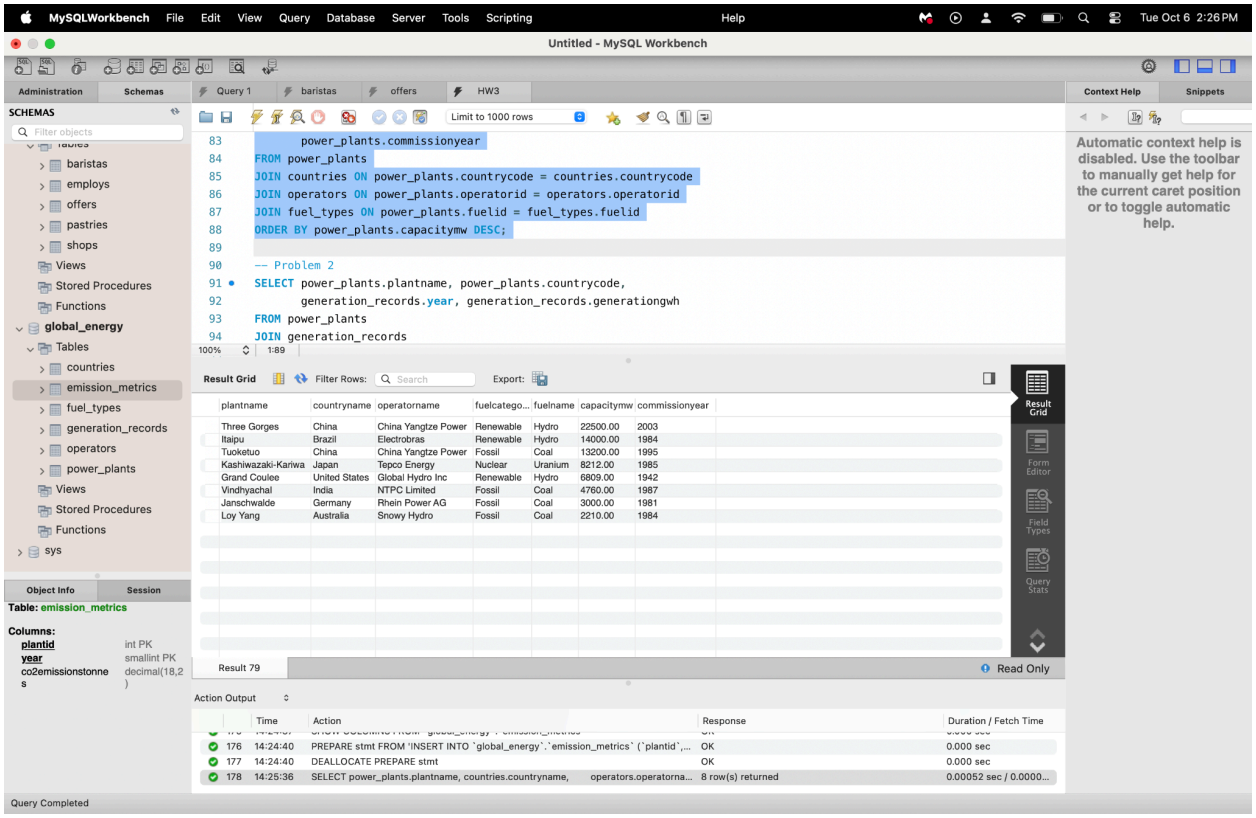
HW3: Global Energy and Power Plants

EER Diagram



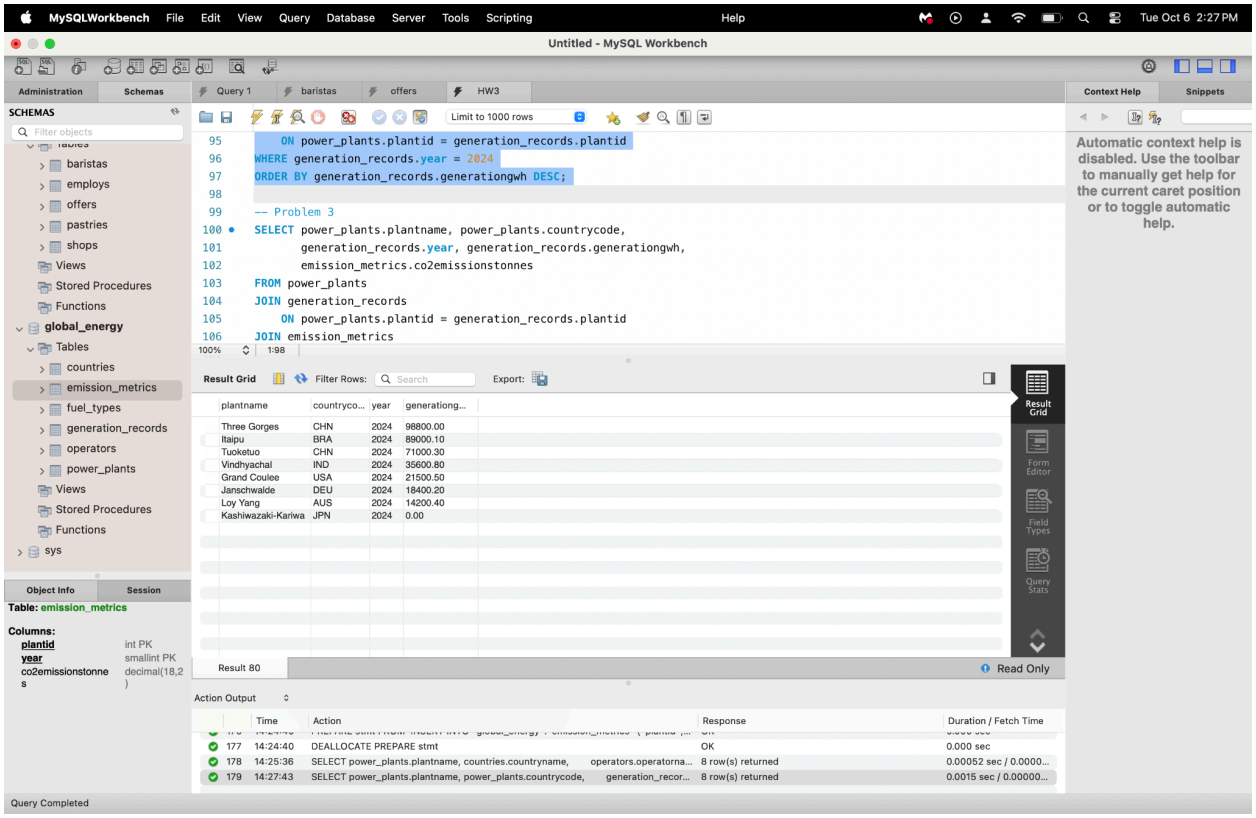
Problem 1

```
SELECT power_plants.plantname, countries.countryname,
operators.operatorsname,fuel_types.fuelcategory,
fuel_types.fuelnsme,power_plants.capacitymw,
power_plants.commissionyear
FROM power_plants
Join countries ON power_plants.countrycode =countries.countrycode
JOIN operators ON power_plants.operatorid = operators.operatorid
JOIN fuel_types ON power_plants.fuelid = fuel_types.fuelid
ORDER BY power_plants.capacitymw DESC;
```



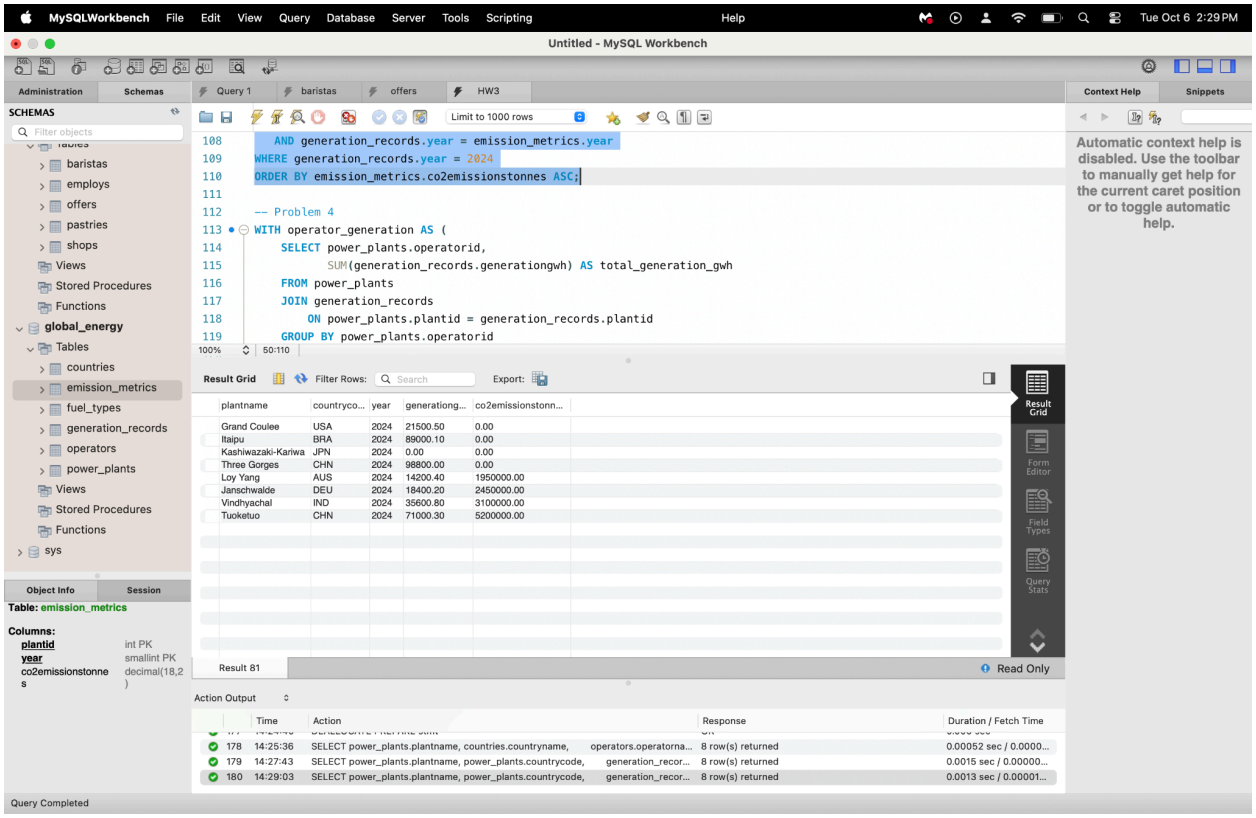
Problem 2

```
SELECT power_plants.plantname, power_plants.countrycode,
       generation_records.year, generation_records.generationgwh
FROM power_plants
JOIN generation_records
   ON power_plants.plantid = generation_records.plantid
WHERE generation_records.year = 2024
ORDER BY generation_records.generationgwh DESC;
```



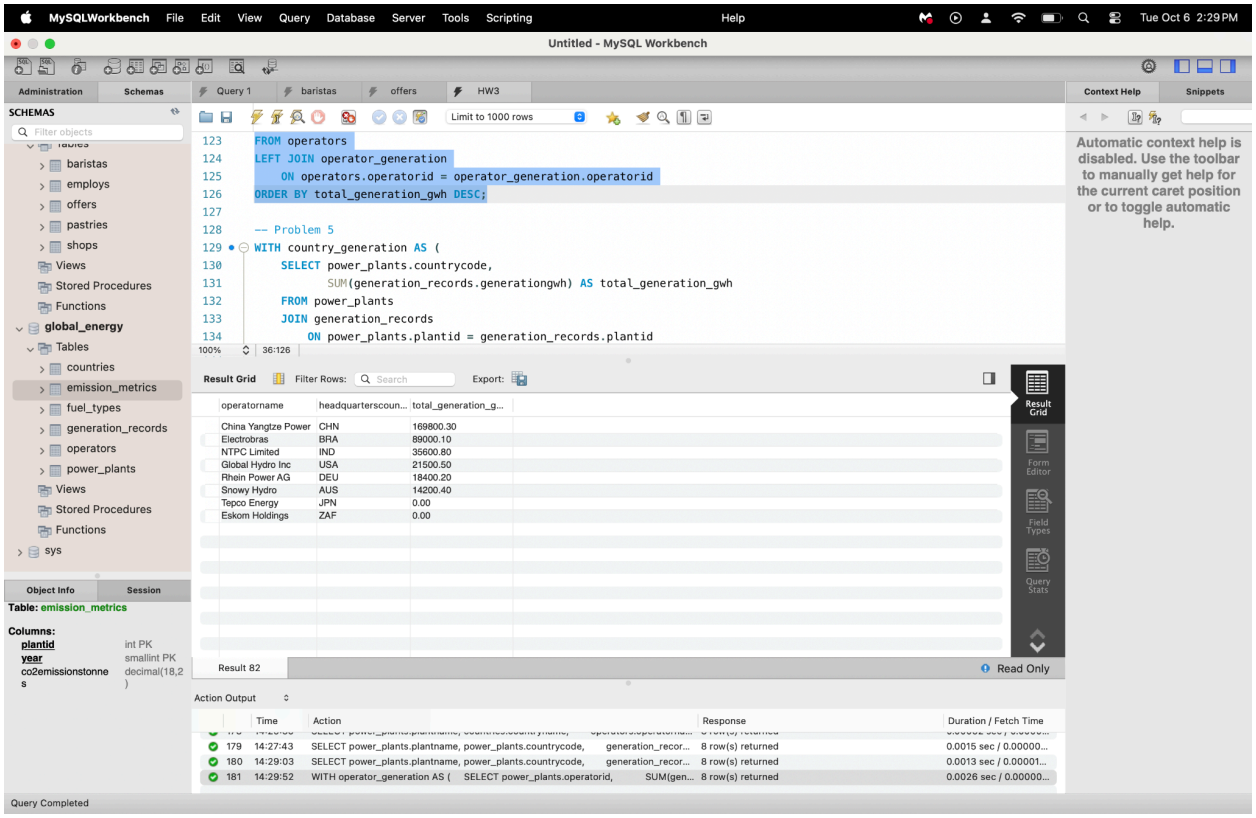
Problem 3

```
SELECT power_plants.plantname, power_plants.countrycode,
       generation_records.year, generation_records.generationgwh,
       emission_metrics.co2emissionstonnes
FROM power_plants
JOIN generation_records
    ON power_plants.plantid = generation_records.plantid
JOIN emission_metrics
    ON generation_records.plantid = emission_metrics.plantid
   AND generation_records.year = emission_metrics.year
WHERE generation_records.year = 2024
ORDER BY emission_metrics.co2emissionstonnes ASC;
```



Problem 4

```
WITH operator_generation AS (
   SELECT power_plants.operatorid,
         SUM(generation_records.generationgwh) AS total_generation_gwh
   FROM power_plants
   JOIN generation_records
      ON power_plants.plantid = generation_records.plantid
   GROUP BY power_plants.operatorid
)
SELECT operators.operatorname, operators.headquarterscountry,
      COALESCE(operator_generation.total_generation_gwh, 0) AS total_generation_gwh
FROM operators
LEFT JOIN operator_generation
   ON operators.operatorid = operator_generation.operatorid
ORDER BY total_generation_gwh DESC;
```



Problem 5

```
WITH country_generation AS (
   SELECT power_plants.countrycode,
         SUM(generation_records.generationgwh) AS total_generation_gwh
   FROM power_plants
   JOIN generation_records
      ON power_plants.plantid = generation_records.plantid
   GROUP BY power_plants.countrycode
),
country_emissions AS (
   SELECT power_plants.countrycode,
         SUM(emission_metrics.co2emissionstonnes) AS total_co2_emissions_tonnes
   FROM power_plants
   JOIN emission_metrics
      ON power_plants.plantid = emission_metrics.plantid
   GROUP BY power_plants.countrycode
)
SELECT countries.countryname,
      COALESCE(country_generation.total_generation_gwh, 0) AS total_generation_gwh,
      COALESCE(country_emissions.total_co2_emissions_tonnes, 0) AS
total_co2_emissions_tonnes
FROM countries
LEFT JOIN country_generation
```

```
    ON countries.countrycode = country_generation.countrycode
LEFT JOIN country_emissions
    ON countries.countrycode = country_emissions.countrycode
ORDER BY total_generation_gwh DESC;
```

